



Linear Transformations · Worksheet 1

Core Pure 1 Chapter 7: Linear Transformations. Answer all questions. Show your working clearly. Total: 64 marks. Time: 80 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 7. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Rotations are anticlockwise about the origin O unless stated otherwise. The transformation “**P** followed by **Q**” is represented by **QP**. Give exact answers where possible.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Matrices as transformations and standard matrices	18	
B Combining transformations and area scale factor	20	
C Invariant points and invariant lines	21	
Roar-level challenge	5	
Total	64	

Section A Matrices as transformations and standard matrices

18 marks

1. The transformation T is represented by the matrix $\begin{pmatrix} 2 & 1 \\ -1 & 4 \end{pmatrix}$. [2]
Find the images under T of the points $(3, -2)$ and $(-1, 1)$.

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2. Describe fully the transformations represented by the matrices [2]

(a) $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ (b) $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

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3. Write down the 2×2 matrix representing each of the following transformations. [3]

- (a) An enlargement, scale factor 3, centre O .
- (b) A reflection in the line $y = -x$.
- (c) A stretch parallel to the x -axis, scale factor 2.

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4. A linear transformation T maps the point $(1, 0)$ to $(3, 1)$ and the point $(0, 1)$ to $(-2, 4)$. [4]

- (a) Write down the matrix representing T .
- (b) Find the coordinates of the point that T maps to $(4, 6)$.

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5. (a) Describe fully the single geometrical transformation represented by the matrix

[2]

$$\mathbf{R} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}.$$

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(b) Hence explain why $\mathbf{R}^6 = \mathbf{I}$.

[1]

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(c) Describe fully the transformation of three-dimensional space represented by

[2]

$$\mathbf{T} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

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(d) Find the image of the point $(2, 3, -1)$ under $\mathbf{T}$.

[1]

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(e) Write down the matrix representing a reflection in the plane $z = 0$.

[1]

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(Total for Question 5 is 7 marks)

**Section B Combining transformations and area scale factor**

20 marks

6. $\mathbf{P} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ represents a reflection in the x -axis and $\mathbf{Q} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ represents a rotation of 90° about O . [2]

Find the matrix representing the transformation $\mathbf{P}$ followed by $\mathbf{Q}$.

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7. For the matrices in Question 6, find the matrix representing $\mathbf{Q}$ followed by $\mathbf{P}$, and describe fully the single transformation it represents. [3]

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8. A triangle T has area 6 cm^2 . It is transformed by the matrix $\mathbf{M} = \begin{pmatrix} 2 & 5 \\ 1 & -1 \end{pmatrix}$. [3]
- Find the area of the image of T , and state what the sign of $\det \mathbf{M}$ tells you about the transformation.

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9. $\mathbf{M} = \begin{pmatrix} k & 4 \\ 1 & k \end{pmatrix}$, where k is a real constant. $\mathbf{M}$ transforms a shape of area 3 square units to a shape of area 15 square units. [4]
- Find the possible values of k .

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10. The transformation U is a stretch parallel to the x -axis with scale factor 3.
The transformation V is a rotation of 90° **clockwise** about O .

(a) Write down the matrices $\mathbf{U}$ and $\mathbf{V}$ representing U and V . [2]

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(b) Find the matrix $\mathbf{W}$ representing U followed by V . [2]

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(c) The triangle with vertices $(1, 1)$, $(3, 1)$ and $(1, 4)$ is transformed by $\mathbf{W}$. Find the area of its image. [2]

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(d) Find $\mathbf{W}^{-1}$ and describe it as a combination of two transformations, in the correct order. [2]

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(Total for Question 10 is 8 marks)



Section C Invariant points and invariant lines

21 marks

11. Show that the point $(4, 2)$ is invariant under the transformation represented by $\begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}$. [2]

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12. Find the equation of the line of invariant points of the transformation represented by $\begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}$. [3]

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13. $\mathbf{M} = \begin{pmatrix} 1 & 1 \\ 0 & 3 \end{pmatrix}$. [3]

Show that the line $y = 2x$ is an invariant line of the transformation represented by $\mathbf{M}$, and explain why it is not a line of invariant points.

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14. Find the equations of the invariant lines through the origin of the transformation represented by $\begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$. [4]

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15. $\mathbf{M} = \frac{1}{5} \begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix}$.

- (a) Find $\det \mathbf{M}$ and explain what it tells you about the effect of the transformation on areas and on orientation. [2]

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- (b) Find the equation of the line of invariant points of the transformation represented by $\mathbf{M}$. [3]

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- (c) Show that every line of the form $y = -2x + c$, where c is a constant, is an invariant line of the transformation. [3]

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- (d) Hence describe fully the transformation represented by $\mathbf{M}$. [1]

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(Total for Question 15 is 9 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. Find **all** the invariant lines of the transformation represented by the matrix $\begin{pmatrix} 1 & 3 \\ 1 & -1 \end{pmatrix}$. [5]

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END OF WORKSHEET

Total: 64 marks